DAY-5 Session

Quantum Applications and Future Directions

Bridging Quantum Mechanics, Machine Learning, and Industrial Optimization for the Next Era of Computing.

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What is Machine Learning?

Learning from data and improving with experience.



The Core Concept

A Branch of AI: Machine Learning is a part of Artificial Intelligence (AI) that enables computers to learn from data.

Pattern Recognition: ML helps computers identify patterns and relationships in data.

Learning from Data: Instead of programming every rule, we train computers using examples and data.

Prediction and Decision-Making: After learning from data, ML models can make predictions or help in making decisions.

Improving Performance: With relevant, high-quality data and proper training, ML models can improve their performance.

Example: An email spam filter learns from examples of spam and genuine emails. It identifies patterns in new emails and predicts whether they are spam.



The 5-Step Operational Process

1

Historical Data

Gather large volumes of relevant, high-quality historical data.

2

Data Preprocessing

Clean, format, and structure the data to remove noise.

3

Model Training

Feed the clean data into designated ML algorithms.

4

Learn Patterns

The system independently maps correlations and hidden structures.

5

Predictions

Deploy the trained model to process new data and execute decisions.

The Three Pillars of Machine Learning



Supervised Learning

The algorithm learns from a set of explicitly labeled data consisting of matched input-output pairs.

Example: Spam Detection



Unsupervised Learning

The algorithm identifies hidden patterns, groupings, or structures within completely unlabeled data.

Example: Customer Segmentation



Reinforcement Learning

The agent learns by interacting dynamically with an environment, maximizing a reward signal over time.

Example: Autonomous Driving

Challenges and Limitations of Classical ML

Why we need to look beyond classical approaches

Classical ML has achieved remarkable global success.

Faces fundamental challenges with complex, large-scale problems.

High-dimensional real-world data exposes current limitations.

Algorithmic & Data Complexities

Curse of Dimensionality

As the number of features increases, data becomes sparse, making it harder for ML models to identify patterns

Complex Feature Spaces

Complex relationships in data require more powerful models, making accurate predictions more difficult.

Optimization Challenges

Finding the best model parameters is difficult because the training error can have a complex landscape.

Processing Massive Data

Processing huge amounts of data requires significant computing power, memory, and time.

Impact on Real-World ML

Slower Time-to-Insight

Delays in training and inference reduce overall operational agility and responsiveness.

Higher Costs & Energy

Increased hardware and maintenance raise massive sustainability concerns.

Limited Scalability

Extremely hard to scale across increasing data sizes, users, and real-time application needs.

Performance Limitations on Certain Complex Problems

Models may fail completely to capture highly complex, underlying patterns in the hardest datasets.

Why Explore Quantum?

Can quantum computing potentially accelerate selected machine-learning and optimization tasks?

- Uses quantum states and quantum operations rather than classical bit states and logic gates.
- Potential to handle exponentially large feature spaces highly efficiently.
- Can mathematically accelerate complex computations and optimization tasks.
- Enables completely new models and approaches infeasible for classical machines.

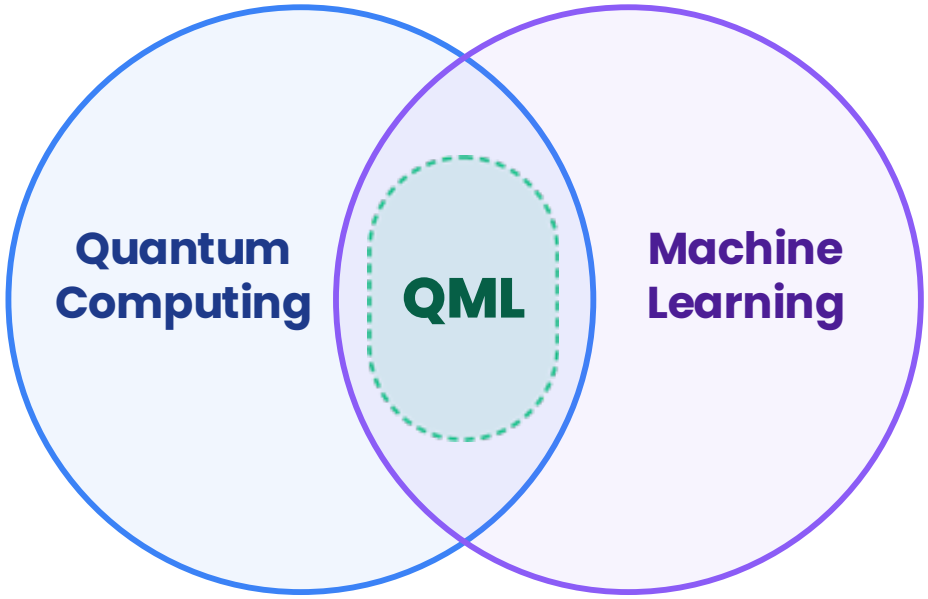
Potential Advantage: Certain quantum algorithms provide theoretical speedups for specific computational problems under specific assumptions.



Introduction to Quantum Machine Learning

Combining quantum computing principles with machine learning to process information in new ways and solve complex problems efficiently.

The Intersection of Quantum Computing and AI



What it is

The convergence of Quantum Computing, Machine Learning, and Optimization to build intelligent systems.

The Objective

To leverage distinct quantum phenomena (superposition, entanglement) to solve complex learning problems more efficiently.

The Philosophy

QML is designed to *enhance*, not completely replace, classical machine learning for specific, computationally intensive tasks.

Why Combine Quantum with ML?

Exponential Speedup

Quantum algorithms can solve certain mathematical and data problems much faster than classical counterparts.

High-Dimensional Spaces

Quantum states naturally represent data in exponentially large feature spaces using fewer computational resources.

Pattern Recognition

Quantum interference and entanglement enable the capture of complex correlations that classical models miss.

What QML Enables & Real-World Impact

High-Dimensional Feature Spaces

Quantum feature maps can embed data into high-dimensional Hilbert spaces whose dimension grows exponentially with the number of qubits.

Potential Data Efficiency

Some QML models may offer advantages in particular low-data or highly structured settings, but consistent generalization advantages have not been established.

Healthcare & Biotech

Accelerating drug discovery through precise, large-scale molecular simulations.

Financial Modeling

Running incredibly complex risk assessment models and portfolio optimizations instantly.

Tech & Cybersecurity

Potentially enhancing selected anomaly-detection tasks and advanced image recognition accuracy.

Core AI Enhancement

Pushing classical Classification, Regression, and Generative Modeling into a new frontier.

The Four Paradigms of Processing

CC

Standard ML
Classical Data on
Classical Devices

CQ

Near-Term QML
Classical Data on
Quantum Devices

QC

Quantum Science
Quantum Data on
Classical Devices

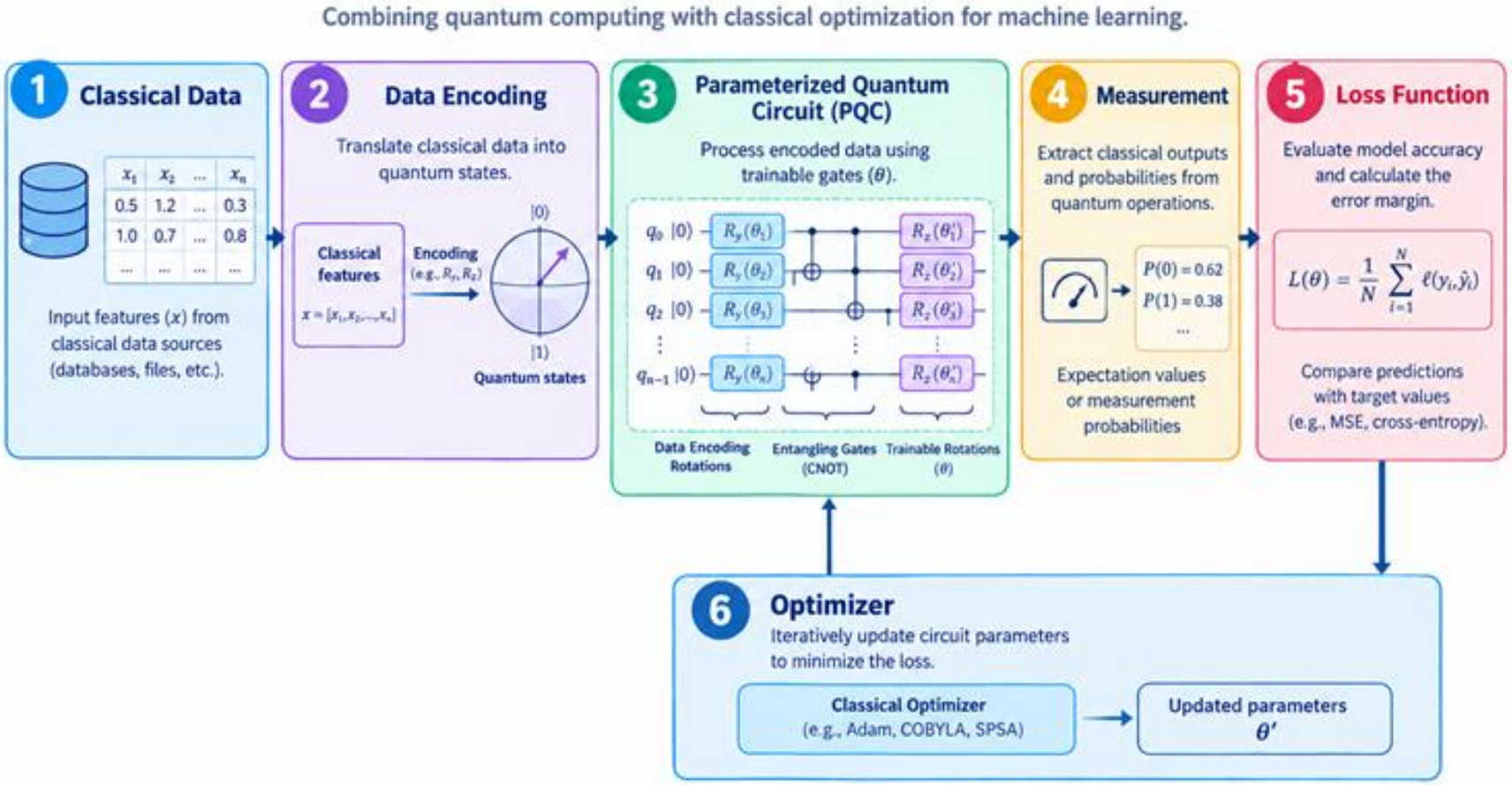
QQ

Long-Term Research Direction
Quantum Data on
Quantum Devices



Block	Data	Processing Device	Description	Example
C for C (CC)	Classical Data	Classical Computer	Traditional machine learning where classical datasets are processed using classical algorithms running on CPUs or GPUs.	Logistic Regression, SVM, Decision Trees, Neural Networks
Q for C (QC)	Quantum Data	Classical Computer	Quantum data generated from quantum experiments or simulations is analyzed and modeled using classical computing techniques.	Analyzing quantum sensor data or quantum experiment measurements
C for Q (CQ)	Classical Data	Quantum Computer	Classical data is encoded into quantum states and processed by a quantum computer to perform machine learning tasks. This is the primary focus of modern Quantum Machine Learning.	Variational Quantum Classifier (VQC), Quantum Kernel Methods (QSVM), Quantum Neural Networks (QNN)
Q for Q (QQ)	Quantum Data	Quantum Computer	Quantum data is processed directly on a quantum computer using quantum algorithms. This is expected to become more important as large-scale quantum hardware matures.	Learning from quantum states, quantum error correction, quantum chemistry data analysis

Hybrid Quantum-Classical Learning



Mapping Classical ML to Quantum

Supervised Learning

Assigning discrete labels or continuous numerical values.

Classification	→	**QSVM**
Estimation	→	**VQR**

Unsupervised Learning

Finding hidden groupings and compressing data states.

Clustering	→	**Quantum K-Means**
Dim. Reduction	→	**Quantum PCA**
Anomaly Detection	→	**Quantum Kernel**

Data Encoding & Quantum Feature Maps

Before processing, classical data must be mapped into a quantum state via a feature map (Hilbert Space Mapping).

The Concept & Separability

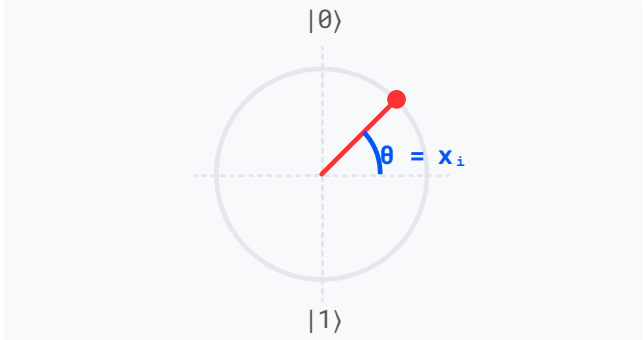
A specialized quantum circuit encodes classical data into a higher-dimensional quantum feature space (Hilbert space).

Highly overlapping, non-linear data points in classical 2D space often become easily separable by a linear hyperplane.



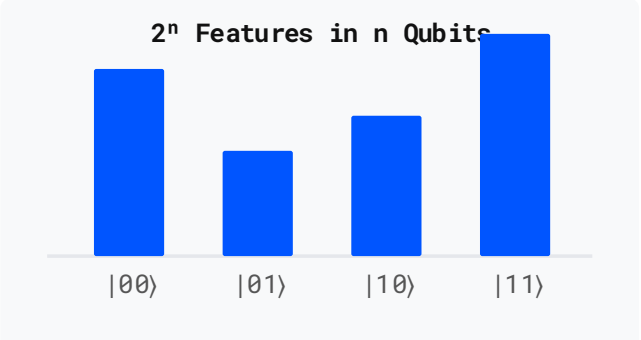
Angle Encoding

Maps features directly to qubit rotation angles (e.g., Ry or Rx gates).



Amplitude Encoding

Embeds a normalized data vector into the amplitudes of a quantum state, offering exponential storage capacity.



Core Quantum ML Algorithms

Quantum Classifier: Uses specialized quantum circuits to classify data, actively leveraging superposition and interference.

Quantum Neural Networks (QNN): The quantum version of classical neural nets, utilizing parameterized quantum circuits for learning.

Variational Quantum Classifier (VQC): A powerful hybrid quantum-classical model where parameters are optimized with classical optimizers.

Quantum Convolutional Neural Networks (QCNN): Extracts hierarchical features using quantum convolutions and reduces qubits via pooling.

Quantum Autoencoders: Learns highly compact quantum representations, exceptionally useful for dimensionality reduction.



Kernels & Optimization Methods

Quantum Support Vector Machine

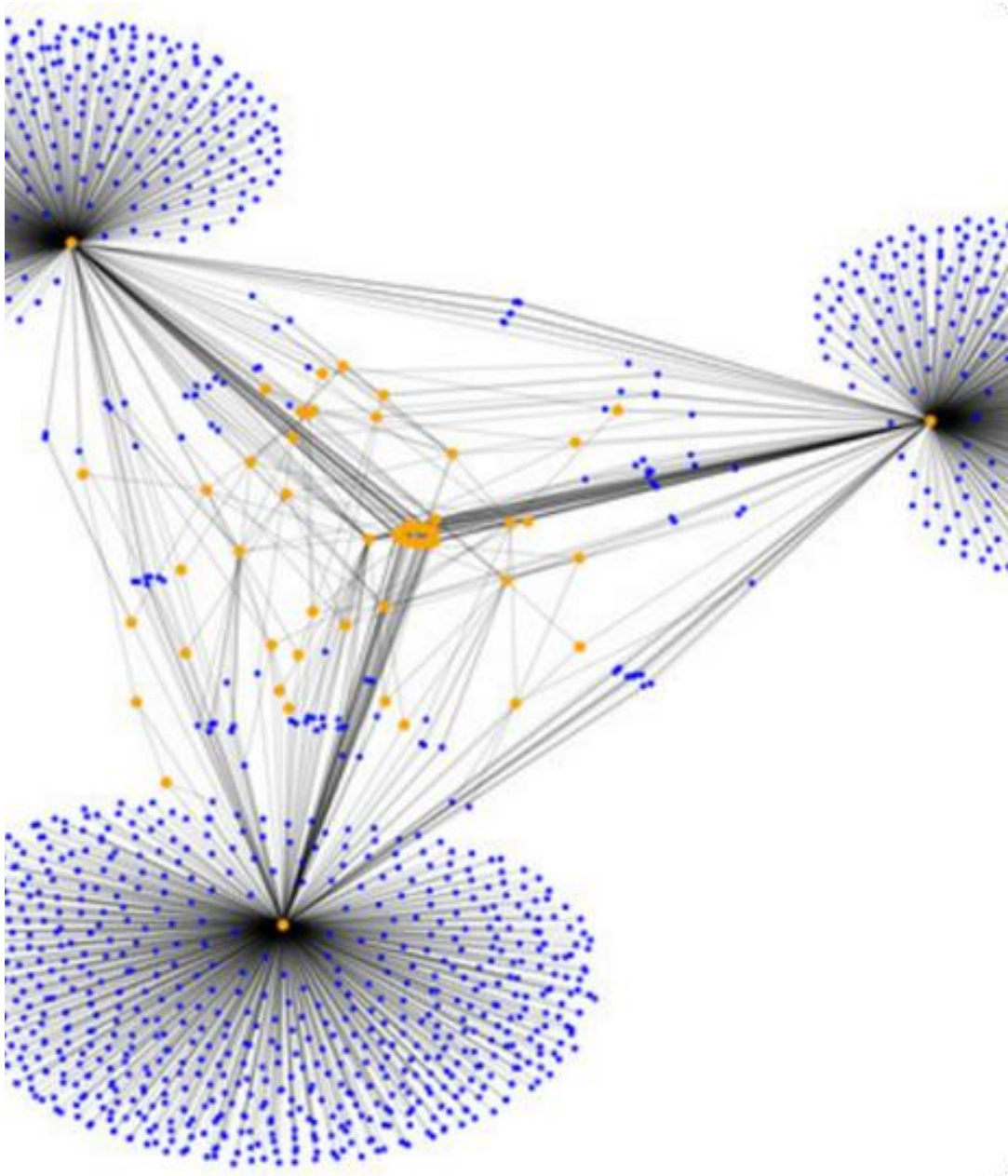
Maps standard data to a quantum feature space, effectively finding optimal hyperplanes using highly complex quantum kernels.

QAOA for Optimization

The Quantum Approximate Optimization Algorithm is specifically designed to solve massive combinatorial optimization problems.

VQE for Learning Problems

The Variational Quantum Eigensolver finds the ground state or optimal operational parameters for complex scientific problems.



Quantum Optimization

Solving the Intractable with Quantum Computing

What is Quantum Optimization?

Quantum optimization applies quantum computing principles aim to solve certain optimization problems more efficiently than classical methods.

It leverages deep computational primitives to navigate massive solution spaces that classical algorithms simply find intractable.

- Quantum optimization algorithms encode many candidate configurations into a quantum state and use quantum interference, problem-specific Hamiltonians and measurement to increase the probability of obtaining high-quality solutions.
- Targets high-value business challenges where incremental gains cascade into vast operational advantages.



Source: https://www.bqpsim.com/quantum-optimization/quantum-optimization-problems

Key Industry Applications

Routing and Logistics Vehicle routing, delivery scheduling, and complex network flow management.

Finance Portfolio optimization, strict risk balancing, and advanced fraud detection models.

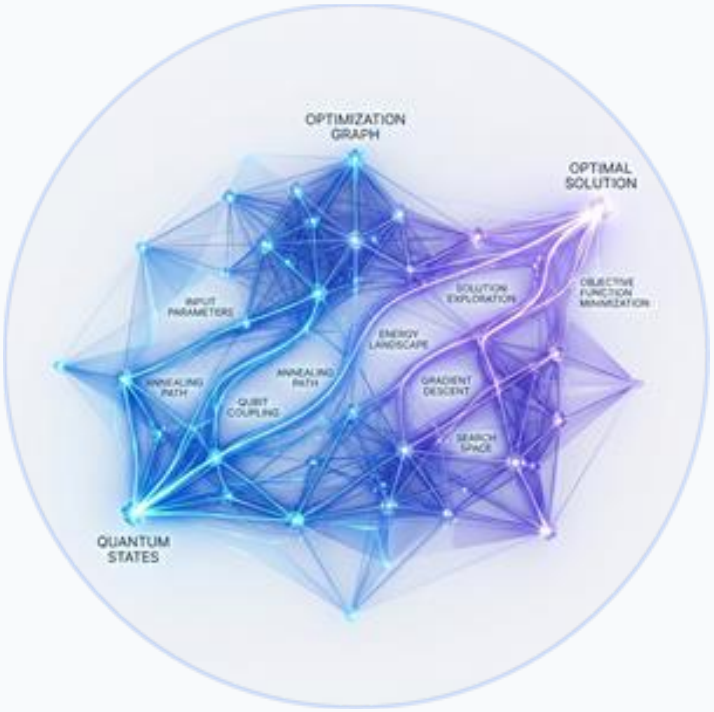
Manufacturing & Energy Production scheduling, supply chain optimization, and efficient grid energy distribution.

Aerospace, Defense & Engineering Mission planning, satellite resource management, and high-dimensional parameter exploration.

Dominant Algorithms

The specialized engines driving practical quantum optimization today.

QAOA: Quantum Approximate Optimization Algorithm



Hybrid Quantum-Classical Design

QAOA is built for near-term noisy quantum hardware. It encodes problems into a cost Hamiltonian, alternating between quantum gate operations (cost and mixer gates) and classical optimization loops to steer parameters toward better solutions.

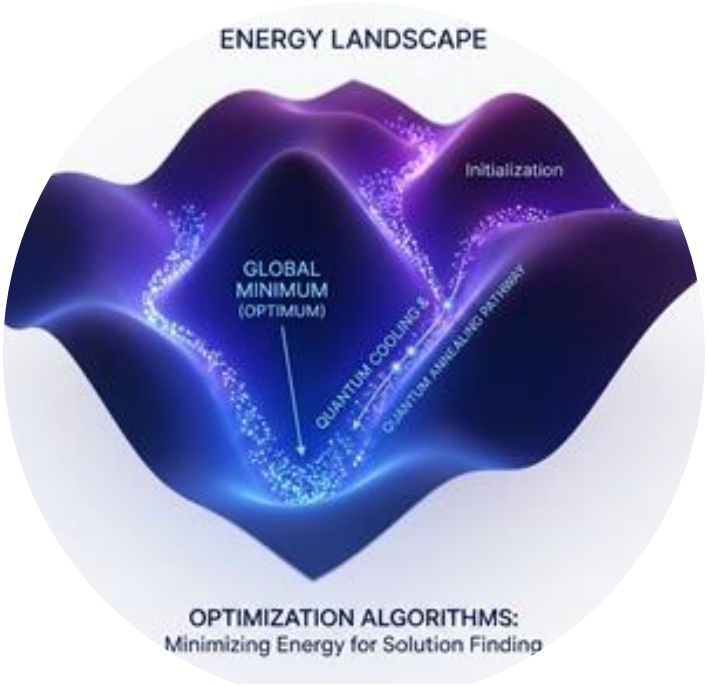
Target Applications

Commonly studied for combinatorial optimization problems such as MaxCut, scheduling, routing and resource allocation such as MaxCut, vehicle routing, job scheduling, resource allocation, and general constraint satisfaction.

Key Strengths

Hardware-platform independent at the algorithmic level, but practical performance is strongly hardware-dependent. Operates seamlessly on gate-model quantum computers (IBM, Google, IonQ) while fully leveraging your existing classical HPC infrastructure.

Quantum Annealing: Physical Optimization Engine



Physical Cooling Process

Uses a physical cooling process to find optimal solutions. The system starts in a high-energy superposition and gradually anneals to its lowest-energy configuration.

How It Works

Encodes problems as an energy landscape. It slowly reduces quantum fluctuations, allowing the system to settle into low-energy states while quantum tunneling helps escape local minima.

Target Applications

Best for large constraint-based variables (logistics, supply chain routing, network design, portfolio construction, manufacturing scheduling).

Key Strength

Highly efficient for energy minimization and scales significantly better to large problem sizes than gate-model QAOA on current NISQ-era hardware.

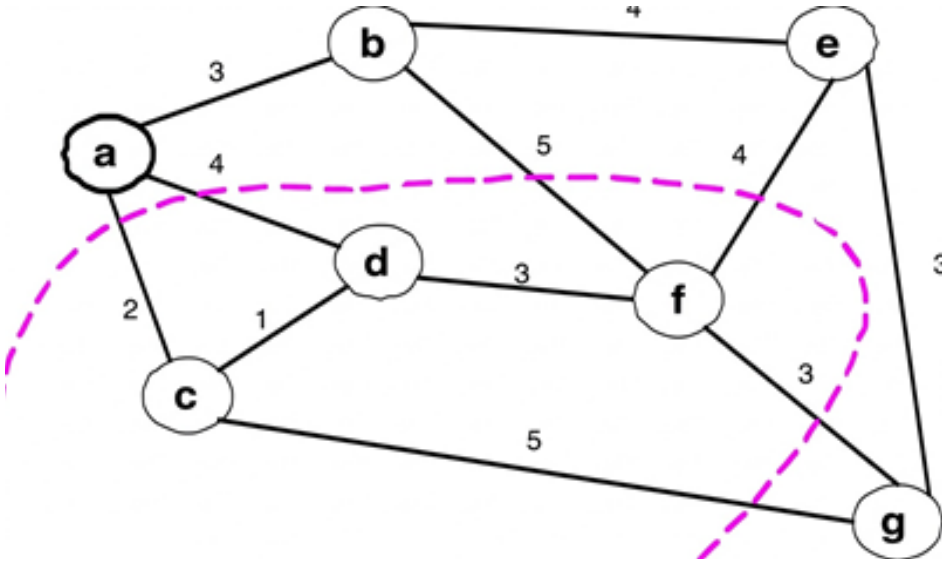
Solving MaxCut with QAOA

From Combinatorial Graphs to Quantum Approximate Optimization

Combinatorial Optimization & The Max-Cut Problem Formulation

The Max-Cut problem is a classic NP-hard combinatorial challenge ideal for quantum optimization.

- **The Objective:** Partition graph vertices into two disjoint sets to maximize the total weight of edges crossing the cut.
- **QUBO Mapping:** The problem is formulated as Quadratic Unconstrained Binary Optimization (QUBO). We assign a binary variable (0 or 1) to each node.
- **Hamiltonian Encoding:** The classical QUBO is transformed into a quantum Hamiltonian. The Pauli-Z operator replaces binary variables, assigning an eigenvalue of +1 or -1. Minimizing this Hamiltonian's energy maximizes the number of cut edges.



The Quantum Approximate Optimization Algorithm (QAOA) Mechanics

QAOA is a hybrid quantum-classical framework designed to find approximate solutions for difficult optimization problems.

01 Alternating Layers

QAOA utilizes a parameterized circuit that alternates between two Hamiltonians in layers (p):

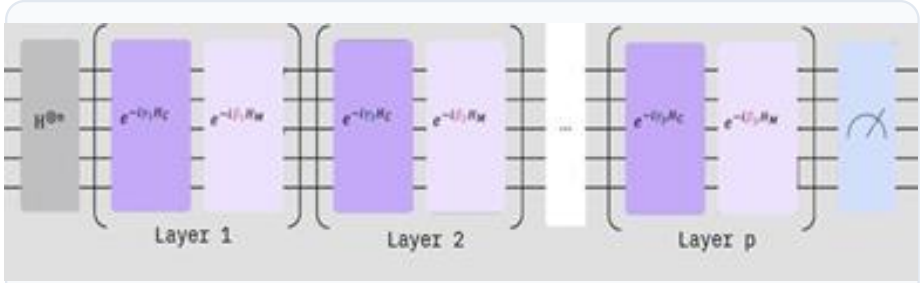
- **Cost Hamiltonian:** Encodes the specific optimization problem, such as maximizing the no. of edges in Max-Cut problem.
- **Mixer Hamiltonian:** Explores candidate solution states by introducing superposition and entanglement.

02 Iterative Optimization

A classical optimizer updates the parameters (γ and β) for each layer to minimize the expectation value of the cost Hamiltonian.

03 Performance

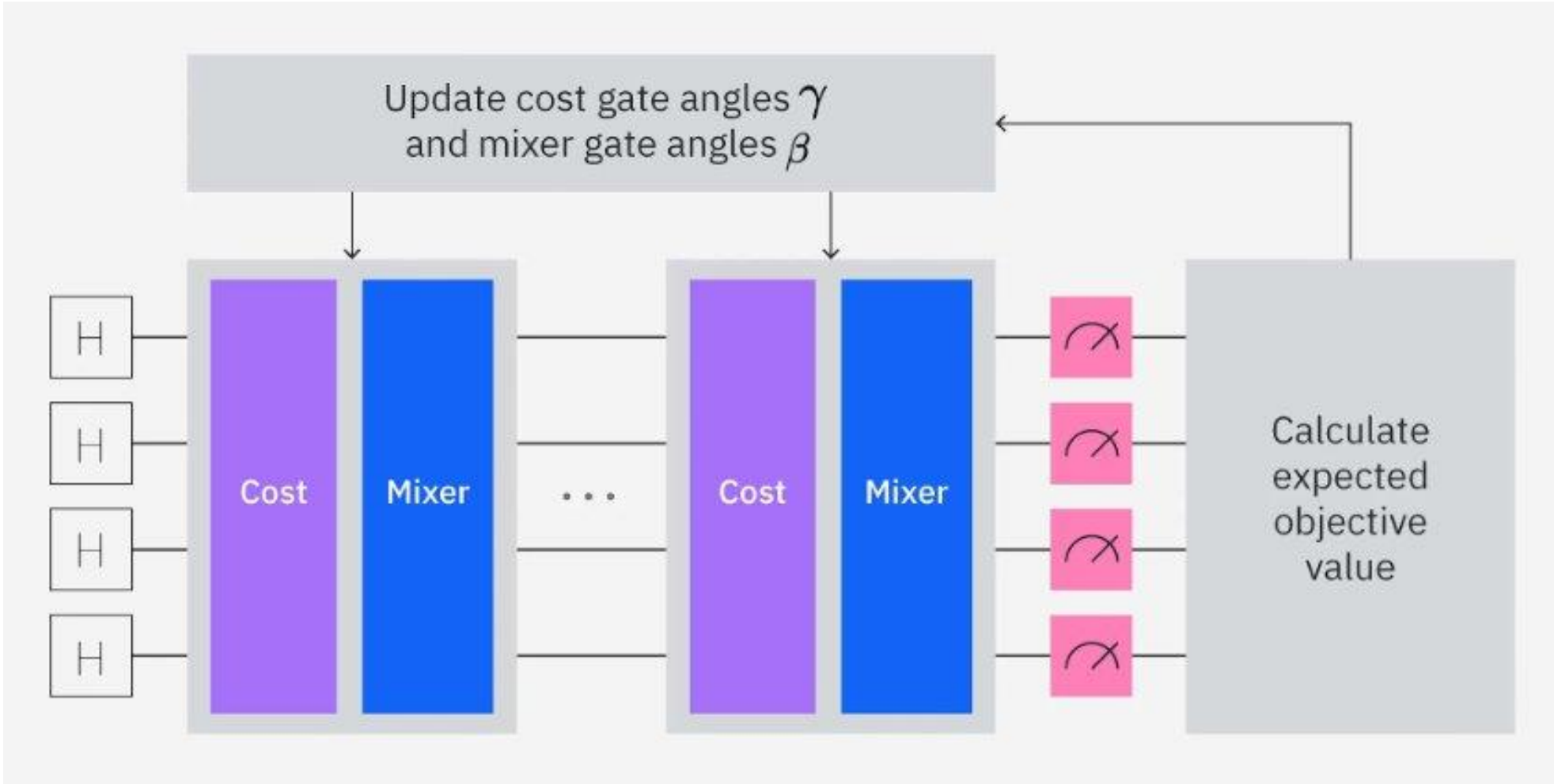
As the number of layers (p) increases, QAOA approaches the adiabatic algorithm, improving accuracy but increasing circuit depth and computational cost.



QAOA Parameterized Ansatz Circuit

Hybrid Quantum-Classical Loop

The algorithm offloads the heavy parameterized state preparation to quantum processors while utilizing classical optimization routines to iteratively tune the parameters (γ, β), mitigating hardware noise in NISQ-era devices.


Update cost gate angles γ
and mixer gate angles β
H
H
H
H
Cost
Mixer
Cost
Mixer
Calculate
expected
objective
value

QAOA Workflow

The step-by-step hybrid quantum-classical process of the Quantum Approximate Optimization Algorithm

01

Define Cost Hamiltonian

Encodes the specific optimization problem into the physical system.

02

Define Mixer Hamiltonian

Enables transition between states to explore the complete solution space.

03

Construct Unitaries

Formulates evolution operators: $\exp(-i\gamma H_C)$ and $\exp(-i\beta H_M)$.

04

Build Quantum Circuit

Prepares initial state and structures it into alternating layers (p).

05

Optimize Parameters

Uses classical algorithms to find optimal values for parameter sets (γ, β).

06

Measure Bitstrings

Executes the circuit on quantum hardware and measures outputs to find the solution.

Case Study: QAOA for Smart Factory Scheduling

Applying QAOA to the FT06 Job Shop Scheduling benchmark to optimize factory operations.

QAOA produced a near-optimal solution (makespan 49 vs. classical baseline 47), demonstrating workflow feasibility rather than quantum advantage.

Problem Scale

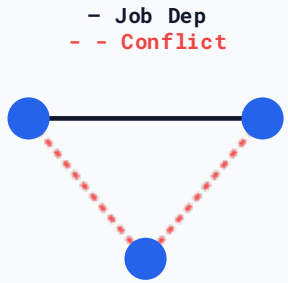
FT06 Job Shop Benchmark

- Scale: 6 jobs, 6 machines, and 36 operations.
- Reduction: Limit to 23 operations for simulation to prevent memory overflow.

Graph Construction

Formulating the QUBO

Operations are mapped as nodes; machine conflicts and job dependencies are edges with weighted penalties.



CLASSICAL GREEDY

47

Baseline makespan performance.

QUANTUM QAOA

49

Near-optimal makespan at depth reps = 4 and 5.

Critical Observation

Depth vs. Optimization Stability

Higher repetitions improved solution space exploration initially, but eventually introduced optimization instability due to increased circuit complexity.

The Variational Quantum Eigensolver (VQE) & Variational Principle

VQE is a hybrid algorithm that estimates the ground state energy of a quantum system.

The Objective

Find the lowest energy state (minimum eigenvalue) of a given Hamiltonian (H), which is computationally intractable for classical machines as systems scale.

The Variational Principle

For any normalized trial state, the expectation value of the Hamiltonian is always greater than or equal to the true ground state energy (E0).

The Strategy

By using a parameterized quantum circuit to prepare trial states and minimizing the expectation value over all parameters, we can approximate the true ground state energy.

The Hybrid VQE Loop

Quantum State Measurement & Classical Optimization

VQE transforms the ground state search into a classical optimization problem over a quantum circuit's parameters.

01 . PREPARE STATE

A parameterized quantum circuit (ansatz) prepares a trial state on the QPU.

02 . MEASURE ENERGY

The system Hamiltonian is decomposed into Pauli strings. The quantum computer measures the expectation value of each Pauli string to compute the total energy.

03 . CLASSICAL OPTIMIZATION

A classical optimizer processes the energy estimate and determines the direction in parameter space that reduces the energy.

04 . UPDATE PARAMETERS

The optimizer updates the circuit parameters and repeats the loop until convergence.

VQE Applications: Quantum Chemistry & Molecular Simulations

APPLICATIONS & SCALING

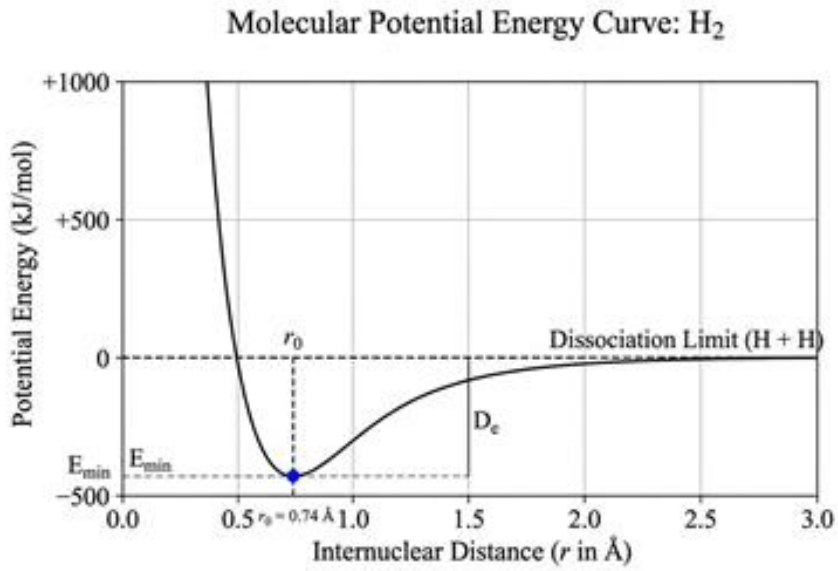
VQE is extensively used to solve complex problems in chemistry and physics.

Molecular Simulations: Calculates molecular ground-state energies for structures like H_2 and LiH, aiding in the discovery of new chemical reaction pathways and materials.

Ansatz Selection: Often utilizes a problem-inspired ansatz, such as Unitary Coupled Cluster Singles and Doubles (UCCSD), which is tailored specifically for quantum chemistry.

Scaling Challenge: Classical methods require memory that grows exponentially with qubits. For example, a 20-qubit system requires a Hamiltonian matrix with roughly one trillion elements, VQE provides a quantum approach for estimating ground-state energies without explicitly storing the full $2^n \times 2^n$ Hamiltonian matrix.

FIGURE 01: POTENTIAL ENERGY DIAGRAM FOR H_2



Key Strengths: Molecular simulation, battery chemistry, materials discovery, and physics-based engineering design where classical simulation hits hard accuracy limits.

Translating the Problem



The Energy Rulebook (Hamiltonian)

To measure energy, we need a mathematical representation of the molecule, called the **Hamiltonian**.

However, quantum computers can't measure the whole complex Hamiltonian at once. Instead, we break it down into smaller, readable puzzle pieces called **Pauli Strings**.

$$H = \sum_{i=1}^{K} c_i P_i$$

We measure each small piece (P_i) individually and sum them up on the classical computer to get the total energy.

The Roadblocks Ahead

While powerful, VQE still faces practical challenges on today's hardware.

Hardware Noise (The NISQ Era) Today's quantum computers are "Noisy Intermediate-Scale Quantum" (NISQ) devices. Environmental interference can cause errors in the qubits, degrading the accuracy of our energy measurements.

Barren Plateaus As problems get larger, the "mountain landscape" we are trying to walk down becomes incredibly flat. The classical optimizer gets stuck because it can't tell which direction is "down."

Measurement Overhead To get an accurate energy score, we can't just measure the quantum circuit once. The circuit may need many repeated measurements (shots) to estimate expectation values with sufficient statistical precision, which takes time and resources.

The Stepping Stone

VQE bridges the gap between near-term, imperfect quantum devices and real-world, complex problems. VQE is an important candidate for demonstrating useful quantum computation in quantum chemistry and materials science, although practical quantum advantage remains an open research question.

Quantum Error Correction

Path to Fault-Tolerant Computing

Protecting fragile logical states from physical-level errors.

Quantum Error Correction (QEC) protects logical quantum information from physical errors by encoding it across multiple physical qubits.

By repeatedly extracting error syndromes, the system detects and corrects physical drift before it causes decoherence, without destroying the quantum superposition.

The Quantum Scalability Problem

Why increasing the number of physical qubits alone is insufficient without quantum error correction and fault-tolerant techniques.

The Fragility of Qubits

01

Hyper-Sensitivity

Qubits are highly sensitive to environmental disturbances, including thermal fluctuations, electromagnetic noise, material defects, and other sources of decoherence. .

02

Non-negligible Error Rates

Physical quantum gates and measurements have error probabilities that are too high for reliable execution of deep quantum circuits without error correction.

03

Error Accumulation

As circuit depth increases, physical errors can accumulate and significantly reduce the fidelity of the computation.

04

The NISQ Era Limit

Hardware-level noise reduction such as cooling, shielding and improved control can reduce physical errors, but scalable fault-tolerant computation requires quantum error correction.

The Anatomy of Quantum Errors

01

Bit-Flip (X Error)

Analogous to classical bit flips, an X error spontaneously flips a qubit from a state of 0 to 1, or vice versa.

$$X|0\rangle = |1\rangle$$

02

Phase-Flip (Z Error)

A Z error changes the relative phase between components of a superposition while leaving computational-basis probabilities unchanged.

$$Z|1\rangle = -|1\rangle$$

03

Depolarization

Depolarizing noise randomly disturbs the quantum state and reduces its distinguishability from the maximally mixed state.

04

Leakage

Leakage occurs when a physical qubit leaves the intended computational subspace $\{|0\rangle,|1\rangle\}$ and occupies another physical energy level.

Why Classical Fails

01

Classical Baseline

Classical error correction adds structured redundancy to data so that errors can be detected and corrected without losing the original information.

02

Quantum Barriers

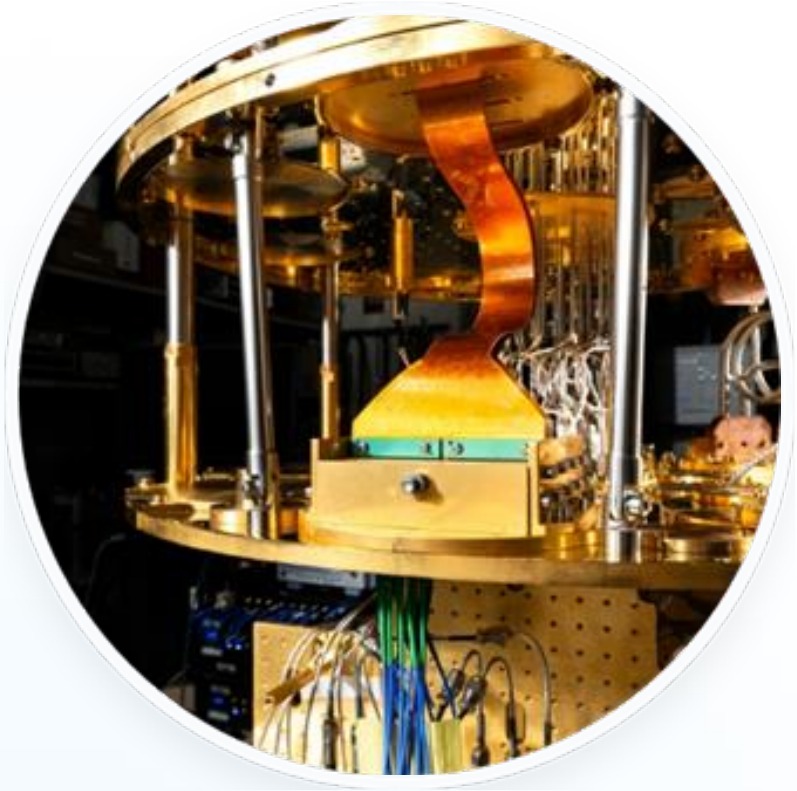
The No-Cloning Theorem prevents making identical copies of an unknown quantum state. Direct measurement of the encoded state can disturb or destroy the quantum information.

03

The Paradigm Shift

Therefore, quantum error correction demands a complete paradigm shift away from classical methodologies.

Redundancy Without Copying



Entanglement & Syndromes

Instead of copying an unknown quantum state, QEC encodes one logical qubit into a multi-qubit code space using structured redundancy and entanglement.

By conducting specialized syndrome measurements on these ancilla (helper) qubits, we detect changes in correlations. The syndrome measurements reveal information about errors without directly measuring the logical quantum information.

Physical-to-Logical Overhead

The Scaling Challenge

Fault-tolerant quantum computing may require many physical qubits per logical qubit. The overhead depends strongly on key variables:

- Error-correcting code design
- Physical & target error rates
- Underlying hardware architecture

Surface-Code Demands

Surface-code implementations represent the current industry baseline but carry immense physical hardware requirements.

Thousands of Qubits

Required per stable logical qubit for demanding workloads.

Many Physical Qubits → One Logical Qubit

Physical Qubits

Noisy hardware qubits that are susceptible to errors.

Logical Qubits

Error-corrected qubits encoded using multiple physical qubits to achieve much lower logical error rates.

The Overhead

Can range from tens to thousands of physical qubits depending on error-correcting code, hardware, and reliability.

The Continuous QEC Cycle

1. Encoding

Encode the logical quantum state into a protected multi-qubit code space using multiple physical qubits.

2. Computation

Perform logical operations using fault-tolerant procedures implemented across the encoded physical qubits.

3. Syndrome Check

Repeatedly measure error syndromes using ancilla qubits to obtain information about physical errors without directly measuring the logical state

4. Decode & Correct

Classical coprocessors analyze syndromes in real-time, the decoder processes syndrome information and determines the appropriate correction or Pauli-frame update.

The Road Ahead

Experiments such as Google's Willow demonstration have shown below-threshold error-correction behavior, an important milestone toward scalable fault-tolerant quantum computing.

Current research focuses on scaling logical qubits, improving physical error rates, reducing decoding latency, and demonstrating increasingly complex fault-tolerant operations.

This is the bridge from experimental science to commercial utility.

Current Quantum Hardware and Future Trends

Evaluating the 2026 Landscape, Technological Breakthroughs, and Strategic Timelines

The 2026 Landscape

Transitioning from Noisy Intermediate-Scale Quantum (NISQ) devices to the foundational era of Fault-Tolerant Architectures.

From NISQ to Fault Tolerance

The NISQ Era (Current Paradigm)

Characterized by limited qubit counts and prohibitive error rates that constrain large-scale computation. The historical bottleneck has been qubit fragility and decoherence. Today, these systems are primarily deployed within hybrid classical-quantum workflows to resolve specific computational bottlenecks.

The Fault-Tolerant Target

The strategic objective for large-scale utility. By relying on real-time error correction, robust logical qubits are formed from clusters of noisy physical qubits. Recent experiments, including Google's 2024 Willow demonstration, have shown below-threshold quantum error correction, strengthening the experimental case for scalable logical qubits.

2026 Modality Comparison

Architecture Type	Scalability & Connectivity	Coherence Time	Leading Developers
Superconducting **High operational speeds; leverages silicon manufacturing.**	Fast operations, complex cryogenic limits	Short (Microseconds)	IBM, Google, Rigetti
Trapped-Ion **Pristine control and exceptionally long coherence times.**	Moderate scalability, pristine control	Very Long (Seconds)	IonQ, Quantinuum
Neutral Atom **Traps uncharged atoms into scalable 2D/3D arrays.**	High (Dense 2D/3D atomic arrays)	Long (Milliseconds)	QuEra, Pasqal, Atom Computing
Photonic **Utilizes light particles; ideal for telecom compatibility.**	Very High, telecom-infrastructure ready	Variable	PsiQuantum, QuiX, Xanadu

Validating Error Correction

Superconducting Breakthroughs

- **IBM Nighthawk:** A 120-qubit processor featuring 218 advanced tunable couplers. It is on track to execute up to 7,500 gates by year-end, targeting early commercial quantum advantage.
- **Google Willow:** A 105-qubit chip that proved the physical scalability of surface code error correction, demonstrating exponential error suppression.
- **IBM Loon:** An experimental prototype utilizing c-couplers to validate the complex architectural elements necessary for full fault tolerance, targeted for 2029.

2.14x

Error Reduction Factor

Exponential Error Suppression

Google's Willow processor demonstrated that scaling a surface code successfully suppressed errors by roughly 2.14x at each step.

This milestone confirms that adding physical qubits extends logical qubit lifespan—proof that fault-tolerant quantum computing is achievable.

Neutral-Atom Breakthroughs

Neutral-atom systems utilize individual atoms held by precise optical tweezers to build highly configurable 3D layouts uniquely suited for complex, rapidly scaling quantum operations.

01 **Strategic Scale Shift:** Neutral atom systems achieved massive milestones in 2026, transitioning from niche academic alternatives to commercial frontrunners in scalability.

02 **Logical Qubit Milestones:** Neutral-atom experiments have demonstrated up to 28 logical qubits encoded in 112 physical atoms in a fault-tolerant Bernstein–Vazirani experiment, while QuEra outlined a direct path to the "Teraquop" regime.

03 **On-Chip Photonics:** Pasqal integrated on-chip silicon photonics to trap individual atoms, bypassing bulky external optics to pave the way for miniaturized 10,000+ qubit systems.



Projected Global PQC Market (2025–2035)



The Cryptographic Threat

As logical qubit scaling accelerates toward a cryptanalytically relevant quantum computer (CRQC), the urgent migration to NIST-standardized Post-Quantum Cryptography (PQC) represents a booming $12.4 billion market opportunity by 2035.

Path to Commercial Fault Tolerance

2026
Surface-code error correction scaling is definitively proven in hardware (Google Willow).

2027
Routine 10,000+ gate execution in hybrid classical-quantum cloud architectures.

2029
IBM and major ecosystem players target the deployment of initial fully fault-tolerant systems.

2030s
PQC migration mandated for critical infrastructure; quantum advantage at broad commercial scale realized.

Scaling via Networked Architectures

$37.5M

NSF HQAN Investment

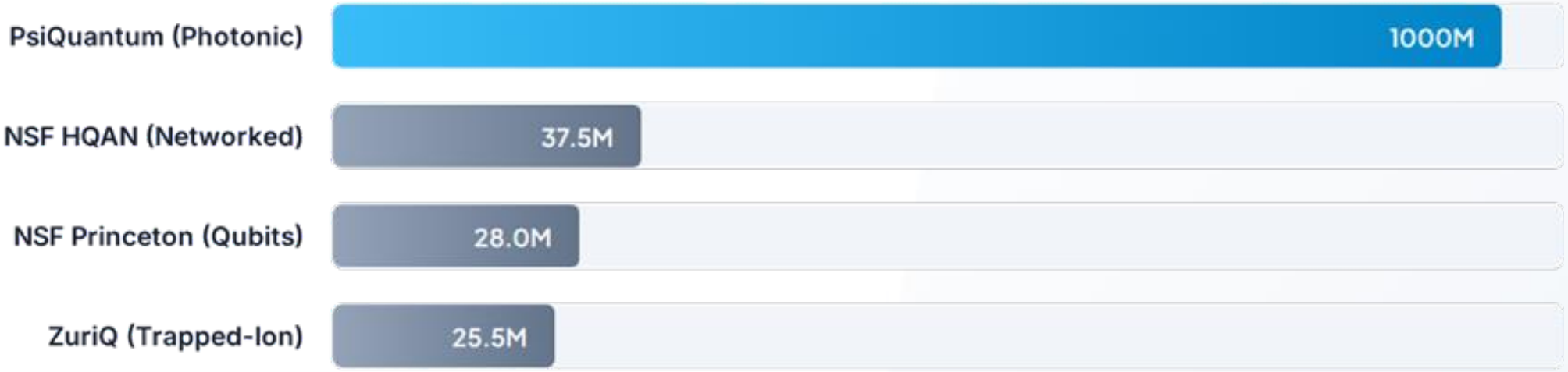
Linking Smaller Processors

The National Science Foundation has heavily invested in the HQAN institute to develop distributed, modular architectures.

Rather than relying on one massive chip, networking smaller, highly reliable quantum processing units (QPUs) is currently gaining industry consensus as the most viable path to overcoming scalability bottlenecks.

Source: quantumnews.in

Recent Quantum Ecosystem Funding (USD Millions)



Capital flow highlights immense confidence in hardware scaling, particularly PsiQuantum's landmark $1B round to support the development of large-scale photonic quantum systems with existing semiconductor infrastructure.

Source: startus-insights.com & quantumzeitgeist.com

Key Hardware Milestones (Aug 2026)

IBM's Scaling Target: Aiming for an unprecedented 100 million quantum operations capacity by 2029 following a strategic deal with HRL Laboratories.

NIST Sensor Advances: Fabricated superconducting photon detectors measuring 0.1 mm, delivering a billion-fold reduction in dark counts.

Error Reduction: D-Wave introduced a new qubit gate that successfully cut operational error rates by a factor of ten.

Supply Chain Security: IonQ acquired SkyWater Technology, securing a U.S.-based quantum chip supply and full vertical integration.

Source: quantumzeitgeist.com & thequbitreport.com

Selected Quantum Hardware and Ecosystem Innovators

IonQ

Trapped-Ion

Building 125 atomic clocks; U.S. chip supply integration

PsiQuantum

Photonic

Scaling systems using established semiconductor foundries

Infleqtion

Neutral Atom

Tracked for 30-qubits; grid vulnerability applications

ORCA Computing

Photonic

Hybrid platforms for advanced AI and machine learning

Hybrid Classical–Quantum Workloads



ORCA Integration

Integrating fiber-optic based quantum processors directly into existing classical systems to accelerate complex machine learning applications and pipeline performance.



NVIDIA CUDA-Q

Leveraging open-source, hybrid programming environments to unify host CPU/GPU systems with QPUs, dramatically scaling machine learning and feature extraction capabilities.



Rigetti TangleLab

Deploying superconducting QPUs directly into classical HPC workflows to actively resolve critical simulation and processing bottlenecks in current enterprise operations.

Future Structural Trends (2026–2030)



Hybrid Compute

Integration of QPUs directly into classical high-performance computing (HPC) environments to offload specific optimization tasks.



PQC Migration

Firms like Atsign and SEALSQ are actively integrating NIST-approved post-quantum cryptography to secure digital infrastructure.



Miniaturization

New fabrication processes, like Imec's EUV nanowires and QTREX's graphene shielding, are driving dense, scalable chip manufacturing.

Source: startus-insights.com & quantumnews.in

The Road to Practical Advantage

2026

Widespread adoption of hybrid classical-quantum models (CUDA-Q) and advanced networked QPU research.

2027

Projected commercial availability of deterministic photonic error correction (Heguang Quantum accelerator).

2029

IBM targets robust execution capabilities of up to 100 million quantum operations.

2030

Major corporate roadmaps (NTT & OptQC) target the realization of 1 million fault-tolerant qubits.

Source: quantumzeitgeist.com & thequbitreport.com

Accelerated Integration: Rigetti & C-DAC

Advancing from Strategic MOU to an $8.4M Hybrid Supercomputing Deployment

A 5-Month Pipeline from Strategic Intent to Hardware Deployment

Phase 1: Exploration & Strategy

September 2025

Memorandum of Understanding (MOU)

- **Context:** Alignment with India's ChipIN initiative to strengthen domestic semiconductor design.
- **Objective:** Explore co-development of hybrid quantum systems and indigenous cryogenic electronics.

5-Month Execution Window

Phase 2: Commitment & Deployment

February 2026

$8.4M Purchase Order Formalized

- **Hardware:** 108-Qubit Quantum Computer utilizing Rigetti's multi-chip architecture.
- **Outcome:** First quantum system actively integrated into C-DAC's existing supercomputing data center.

Sources: cdac.in | quantumzeitgeist.com

Engineering the Hybrid Quantum-Classical Infrastructure



Classical Environment

C-DAC Bengaluru Supercomputing Data Center

Role: Nodal agency for the Government of India's ChipIN initiative and high-performance computing (HPC) leadership.

Delivery Targeted

H2 2026

Seamless Hybrid Integration

Secure On-Premises

Deployment



Quantum Environment

Rigetti 108-Qubit System

Architecture: Proprietary superconducting chiplet-based design.

Edge: Fast gate speeds (50-70ns), enabling high-efficiency workloads.

Thank you!